5. *Anomaly handling*: If a subsystem degrades, FDIR triggers safe modes (loiter, return, or land) while attempting recovery; redundant systems maintain mission continuity where possible.
6. *Recovery and post-mission*: Secure data offload, maintenance diagnostics, and health logs feed back into maintenance scheduling and system learning loops.

4.4 Applications of Third-generation UAV Systems

The third generation of UAV systems represents a transformative phase in unmanned aviation where drones have evolved from specialized reconnaissance tools into multi-domain, mission-capable aerial platforms serving defense, scientific, commercial, and humanitarian purposes. Their advanced autonomy, endurance, and global communication capabilities enabled through satellite networking, precision navigation, and multi-sensor integration broadened the operational spectrum of UAVs across both military and civilian sectors. These applications reflect the growing role of drones in modern warfare, strategic intelligence, environmental monitoring, and datadriven industries.

4.4.1 Military and defense applications

Third-generation drones revolutionized defense operations by introducing real-time surveillance, precision strike, and persistent reconnaissance capabilities. Systems such as the MQ-1 Predator, MQ-9 Reaper, and RQ-4 Global Hawk became indispensable in network-centric warfare, providing a continuous information link between aerial, ground, and command units.

- *Intelligence, surveillance, and reconnaissance (ISR)*: UAVs equipped with EO/IR cameras, SAR radar, and SIGINT systems provided